

Notes: TU IOE PYQ 2070–2081 (17 papers). Regular = **bold**, Back / New Back = normal. Single common paper for BEX + BCT — no programme split. Marks in (); unique values only. Keys: (I)=Importance/Need (+)=Advantage (-)=Disadvantage +Fig=Figure +Eg=Example Vs=Compare/Differentiate.

1. Introduction

- Data mining +process / general steps; applications (2+3)(2+6)(1+5)(4)(10)
- KDD steps +Eg (2+5)(3+7)
- “Data rich but information poor” — justify (8)
- Data warehousing +Eg where used; key features (2+4)(2+5)
- DW architecture +analytical processing (8)
- DW Vs data mining; DW Vs database (similar too) (3+7)(2+2)
- DM system integrated with DB/DW +Fig process (4+2)
- Data warehouse Vs data mart; snowflake scheme +Eg (2+4)(15)

2. Data Preprocessing

- Data types of attributes (10)
- Data pre-processing (I) +major tasks; data clearing (2+7)(4+4)(5+5)
- Data quality elements; data transformation by normalization +Eg (2+6)
- Normalize 200/300/400/600/1000 — min-max (0,1), Z-score, decimal scaling (2+2+2)
- Missing values — methods to handle (5)(2)
- Data integration issues (5)
- Curse of dimensionality +avoid; noisy data impact (5)
- PCA for dimensionality reduction; principal components + % variance (3 × 3 covariance) (10)(4)
- Short notes: data smoothing, data normalization, data transformation (3)(4)

OLAP & Multidimensional

- OLAP operations over multidim. DW; OLAP Vs OLTP (6+4)(10)(5+5)(2+5+3)
- Dimensional data; base & apex cuboid; slice & dice; roll up/down +Eg (2+3+3+3)
- Star schema (spectator/location/game, charge) + OLAP ops (3+3)
- Snowflake schema (time/location/supplier/brand/product) + OLAP ops (3+3)
- Short notes: OLAP; OLAP operations; OLAP and OLTP (4)

Similarity Measures

- Similarity/dissimilarity; cosine + Euclidean (Size/Weight/Color/Taste table) (3+3+3)
- Types of similarity measures +application areas (3)
- Jaccard, SMC, symmetric dissimilarity (Ram/Laxmi/Shyam); cosine of d1,d2 (2+2+2+1)
- Manhattan & Supremum distance matrix; cosine D1/D2; Jaccard+SMC P/Q (2+1+2)
- Minkowski distance — metric properties (10)(4)(3)

3. Classification

- Classification stages +Fig block; precision Vs recall inverse (2+3+5)
- Classification as supervised learning; impurity measures (10)

Decision Tree

- ID3 tree (Age/Income/Student/Credit_Rating → Buy's Computer) (10)
- Information gain tree (Gender/Car/Travel cost/Income → Transport) +predict (8)
- Mushroom (Shape/Color/Odor) — ID3 root +full tree (2+5)
- Decision tree + information gain for attribute selection +Eg (8)
- Gini index +Eg; overfitting in DT (6+4)(10)
- Best root attribute; classifier accuracy (4+6)
- Hunt's algorithm + test conditions per attribute type (10)
- DT with Naive Bayes concept +Eg (10)

Rule Based

- Rule based classifier working +Eg; classification principles (7)(2+8)
- CN2 algorithm; “Accuracy” & “Laplace” rule-evaluation measures (9)
- Conflict resolution strategies (I) (2+4)

Nearest Neighbor

- KNN K=3 Euclidean (iris Sepal/Petal table) (3+6)
- NN classifier + issues + better classifier (1+3+4)

Bayesian

- Naive Bayesian classification +Eg; algorithmic steps +strengths (8)(6)(3+7)
- Prior Vs posterior probability (3)
- Classify X=(Home Owner, Marital Status, Income \$120K) — defaulted borrower (6)
- Classify X=(youth, medium, student yes, fair) — buys computer (RID table) (10)
- Naive Bayes limitation; Bayesian Belief Network +Fig (heart disease CPTs) (3+5)(10)
- Laplacian correction (4)

ANN Classifier

- How ANN / neural net classifier works +algorithm +general considerations (8)(2+6)(5)
- Feed-forward net +Fig — activation at h_1 ; $\partial E/\partial o$; $\partial E/\partial w_{12}$ (2+3+4)

Issues: Overfitting / Validation

- Confusion matrix → accuracy, sensitivity, specificity, precision, recall (10)(4)
- When accuracy fails +Eg; classification error, false alarm rate (3+5)
- ROC +TPR, FPR, precision (1+3+6)
- Confusion matrix from predicted sequences (classifiers X, Y) (10)
- Overfitting problem (4)

4. Association Analysis

- Association analysis (I) +market-basket; where applicable (2+5)(2+6)(10)
- Support & confidence (I) (10)
- Pattern evaluation (I) — statistical interestingness measures +Eg (8)
- Market basket analysis (4)

Apriori

- Apriori principle +Eg; optimizing brute force; generate rules (2+6)(10)
- Why 100% confidence uninteresting but 99% interesting (4)
- Apriori numerals (2+7)(4+5)(8)(12)(10)
 - min sup count 4, conf 80% (T100–T106, F/A/C/D/G...)
 - min sup 2 (10=ACD, 20=BCE, 30=ABCE, 40=BE)
 - sup 50%, conf 80% (6 txns A,B/A,D/A,C/B,E/B,D,E/A,E,C)
 - sup 2/9, conf 7/9 (fast food Order:1–9, M1–M5) +rules $X \wedge Y \rightarrow Z$
 - conf 60%, sup 30%, L=3 +F-List (T1–T6, A,B,C,T,M,P,D,K)
 - sup 20%, conf 80% (TID 1–9, M1–M5)
 - sup 50% (1=ACD, 2=BD, 3=ABCE, 4=BDF)

FP-Growth / FP-Tree

- FP-Growth algorithm +Eg; (+) over Apriori; frequent itemset (2+5)(8)(10)(2+6)(1+8)
- FP-Tree construction numerals (4+4)(2+7)(2+8)
 - TID 01–05 (f,a,c,d,g,i,m,p ...)
 - min sup 60%, conf 80% (T100–T500, {M,O,N,K,E,Y}...)
 - sup 33.34%, conf 60% (HotDogs/Buns/Ketchup/Coke/Chips)
 - Camera/Laptop/Pen drive/Mobile/Earphone (T1–T6)
 - sup 30%, conf 75% (1–8, P,Q,R,S,T)

Categorical / Sequential / Subgraph

- Categorical attributes — handling +issues +Eg (3+3)(2+3+3)(4)
- Subsequences of $\langle \{2,3,5\}\{6,7,8\}\{9,1\}\{7,4\} \rangle$; 4-subsequences of $\langle \{1,3\}\{2\}\{2,3\}\{4\} \rangle$ (3+3)
- Subgraph pattern; candidate sub-graphs by edge-growing +Fig (2)(3+3)
- AprioriALL sequential pattern mining; sequential pattern (3)

5. Cluster Analysis

- Cluster analysis + applications + types of clusters; Vs classification (8)(5)(2+6)(3+4)
- Clustering as unsupervised learning; Bisecting K-means (10)
- Types of clustering methods; partition based methods (10)
- Continuous cluster + algorithm (10)
- Internet-user segmentation — choose algorithm +validate cluster (2+6)

K-means

- K-means algorithm +limitation/demerits; optimal K strategy (2+6)(4+4)(4+6)

- Hierarchical Vs partitioning clustering (2+8)
- K-means numericals (2+8)(5+3)(10)(8)
 - K=2, 6 instances (1,2)(2.5,1)(3.5,1.5)(4,1)(3.5,2.5)(5,3)
 - K=2, A/B (1,2)(2.5,4.5)(4,6)(3.5,4)(4.5,5)(3,6)
 - A/B (1,1)(1.5,2)(3,4)(5,7)(3.5,5)(4.5,5)(3.5,4.5)
 - K=3, matrix X (10 pts), centers (6.2,3.2)(6.6,3.7)(6.5,3.0)

Hierarchical

- Hierarchical clustering +dendrogram; agglomerative Vs divisive +Eg (2+7)(10)
- Dendrogram numericals (8)(4+4)
 - p1–p6 distance matrix (0.24/0.22/0.37...)
 - complete-linkage, Euclidean, p1–p6 (x,y) coordinates
 - A–F distance matrix — single-link & complete-link

DBSCAN

- DBSCAN algorithm +Eg; core/border/noise points; handling noise; avoid issues (3+5)(8)(8+2)(7)
- Neighborhood quantification; Eps & MinPts empirically; Eps1<Eps2 subset proof (2+2+2)
- Density based cluster + algorithm; outliers (8)
- Density reachable / density connected (4)

Issues

- Cluster validity measures; cluster evaluation; issues in clustering

(10)(4)(3)

6. Anomaly / Fraud Detection

- Anomaly detection (I) +distance based method (2+3)(8)(2+6)(4+3)(10)(5)
- Types of anomaly detection schemes; compare three methods (2+2+4)(6)
- Where applicable; importance in security (3+3)(5)
- Types of outlier +Eg; density based outlier detection (5+3)
- Issues in anomaly detection (2)(4)
- False alarm rate & detection rate (P=0.01 / 0.99, 99% normal) (3)
- Nearest-Neighbor for anomaly detection (10)
- Clustering for anomaly detection; base rate fallacy; convex hull method (3)(15)

7. Advanced Applications

- Web mining — structure/taxonomy/categories; challenges (3+5)(6)(8)(5)
- Text mining key steps; page rank +algorithm (4+4)(5)(4)
- PageRank numerical — damping 0.85, 5 iterations, initial 0.25 +Fig (8)
- Multimedia mining; visual data mining (6)(3)(4)
- Time series data mining +Eg (6)(5)(4)